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Variable Significance Testing for the Deep Cox Model

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ABSTRACT
Deep learning has become enormously popular in the analysis of complex data, including event time measurements with censoring. To date, deep survival methods have mainly focused on prediction. Such methods are scarcely used in matters of statistical inference such as hypothesis testing. Due to their black-box nature, deep-learned outcomes lack interpretability which limits their use for decision-making in biomedical applications. This article provides estimation and inference methods for the nonparametric Cox model—a flexible family of models with a nonparametric link function to avoid model misspecification. Here we assume the nonparametric link function is modeled via a deep neural network. To perform statistical inference, we use sample splitting and cross-fitting procedures to get neural network estimators and construct test statistic. These procedures enable us to propose a new significance test to examine the association of certain covariates with event times. We establish convergence rates of the neural network estimators, and show that deep learning can overcome the curse of dimensionality in nonparametric regression by learning to exploit low-dimensional structures underlying the data. In addition, we show that our test statistic converges to a normal distribution under the null hypothesis and establish its consistency, in terms of the Type II error, under the alternative hypothesis. Numerical simulations and a real data application demonstrate the usefulness of the proposed test. Supplementary materials for this article are available online.

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Censored survival data; Curse of dimensionality; Interpretability; Neural networks; Survival analysis

1. Introduction

In the era of "big data," the use of deep learning has become increasingly common for survival analysis with right censored data (Faraggi and Simon 1995; Katzman et al. 2018; Lee et al. 2018). Deep survival models often achieve superior predictive accuracy through improved estimation of the survival function facilitated by flexible representation learning. However, existing approaches are not equipped to test the significance of a covariate's effect on the event time. This article fills this gap by developing a new test of significance for the null hypothesis of no association between one or more specific covariates and the event time.

Survival analysis studies the analysis of event times (i.e., *survival* times) data, and constitutes a major application of statistics in biomedical studies. Specific event times of interest include time to death, disease recovery, new employment, or equipment failure. A key feature of survival data is that there are often incomplete observations due to right censoring. For example, we cannot record the exact event times for those who were lost to follow-up during the period of a study or whose events have not occurred yet by the end of the study. As a result, the (unobserved) event time is larger than the (recorded) observed time for those *right* censored subjects, and traditional regression methods must be adjusted to account for such right-censoring. For a comprehensive treatment on survival analysis, see the monographs by Cox and Oakes (1984) and Fleming and Harrington (1991).

The principal goal of survival analysis is to understand the relationship between an event time and covariates of interest, which often include certain treatments. For example, in a clinical study of 686 patients with node positive breast cancer, it is of interest to determine whether hormone treatment is able to reduce the risk of cancer recurrence following surgery (Sauerbrei et al. 2000). To help answer such questions, classical methods often assume that the cumulative distribution function of the event time has a parametric or semiparametric form, such as the: Cox regression (Cox 1972), accelerated failure time (Kalbfleisch and Prentice 1980; Wei 1992), and proportional odds models (Murphy, Rossini, and van der Vaart 1997). These models include the parameter associated with each covariate, which is used to draw inferences about the effect of this covariate. Among existing survival models, the Cox regression model (Cox 1972) is most widely adopted in applications due to its elegance and interpretibility. It assumes that the conditional hazard function at time u, for a subject with covariate V, takes the form

$$\lambda(u|V) = \lambda_0(u)e^{\beta^\top V}, \tag{1}$$

where β is an unknown parameter to specify the effect of the covariate V while the baseline hazard function λ_0 is an unknown

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nonparametric component. Here a hazard function at time u represents the instantaneous rate of an event at time u given that the event has not yet occurred before u. The successful development of the Cox regression model (Cox 1972) led to significant test of the covariates of interest, say X, through testing the nullity of the components in β that are associated with X. For example, Cox (1972, 1975) estimate β by maximizing the partial likelihood function. Subsequent work by Andersen and Gill (1982) and Begun et al. (1983) established the asymptotic properties of these maximum partial likelihood estimators, demonstrating that they achieve semiparametric efficiency. However, the resulting estimator of β may be biased if the linearity assumption fails to hold.

To avoid model misspecification which is likely under the restriction of linearity, Therneau, Grambsch, and Fleming (1990), Fleming and Harrington (1991), Sasieni (1992), and Dabrowska (1997) replaced the linear predictor $\beta^\top V$ in the Cox regression model (1) by a partially linear structure $\theta^\top X + g(Z)$ with covariate $V = (X, Z)$, an unspecified parameter θ and a nonparametric function g. Although the partially linear model is more flexible in capturing the relationship between Z and the event times, estimating the nonparametric component g from limited data is challenging. Often accomplished through a smoothing method like kernel or spline type smoothing, the resulting estimates suffer from poor convergence rates. A remedy is to impose a low dimensional structure on g, as done by Huang (1999), who assumed that the nonparametric function g has an additive structure without interaction terms and proposed an estimate that has one-dimensional nonparametric convergence rate. Recently, Zhong, Mueller, and Wang (2022) approached the estimation of the function g by a neural network, whose ability to detect low-dimensional structure in high-dimensional space circumvents the need to assume a dimension reduction model in order to avoid the curse of dimensionality. More recently, Sun et al. (2023) adopted the work of Zhong, Mueller, and Wang (2022) to study CT scans of lung cancer patients with penalized high-dimensional linear predictors X. However, these methods failed to account for interactions between covariates X and Z, which may be complex and diminish power in tests of significance if the model is mis-specified.

To address the above concerns, we propose a new test based on deep learning for right censored survival data under a fully nonparametric Cox model (Hastie and Tibshirani 1990; Sleeper and Harrington 1990; O'Sullivan 1993). We consider models where the linear predictor $\beta^\top V$ in the Cox regression model (Cox 1972) is substituted by a multivariate unknown function $g(V)$. We are interested in whether the covariate X influences the survival function, which is equivalent to testing whether the function g depends on X at all. This requires us to estimate the multivariate function g first, which is intractable with kernel/spline-type methods when the dimension of V is large. Driven by the powerful representation-learning capabilities of deep learning (LeCun, Bengio, and Hinton 2015), we leverage neural networks to estimate the function g, allowing the structure, especially the depth, of the neural networks to grow with the sample size n.

Specifically, we randomly split the given survival data into $\mathcal{K}$ disjoint subsets of approximately equal size, where $\mathcal{K}$ is typically chosen to be 5 or 10. The complement of the kth subset is used to get neural network estimates of the function g under the null and alternative hypothesizes, respectively, and the kth subset is used to develop the test statistic by taking the difference between the logarithm of the partial likelihood functions under the null and alternative hypothesizes with their associated g estimates from its complement. Then we employ the cross-fitting method and take the average of the $\mathcal{K}$ test statistics as our test statistic for the hypothesis testing. We further provide theoretical support for the proposed method by establishing the convergence rate of the neural network estimates and asymptotic normality of the test statistic under the null hypothesis and consistency results under the alternative hypothesis. This accomplishes statistical inference for a fully nonparametric Cox model.

This work relates to the studies of Cai, Lei, and Roeder (2022), Dai, Shen, and Pan (2022), and Lundborg et al. (2024), who primarily test the significance of a variable in nonparametric regression models. While sharing the goal of significance testing, our approach differs fundamentally due to two critical aspects: the presence of random censorship inherent in survival data and our specific implementation of cross-fitting. First, the theoretical frameworks in their works follow classical results for sums of iid random variables. In contrast, the log partial likelihood in survival context introduces complex dependency within the at-risk set. This dependency, compounded by random censorship, requires additional tools from empirical process theory to tackle the theory. Second, while they use a simple data splitting method (estimation set vs. inference set), we employ cross-fitting to construct our test statistic in order to gain efficiency. The censoring mechanism induces additional dependence structures that are not accounted for in typical iid cross-fitting frameworks, so the involved theory is more complicated. Furthermore, our simulation studies demonstrate that cross-fitting significantly increase the power of the test compared to the single-splitting approaches in this censored data setting. An additional distinction is our need to establish the convergence rates of neural network estimators under random censorship, a key prerequisite for our inference analysis, whereas Dai, Shen, and Pan (2022) assumes such rates based on existing literature for standard regression settings.

Similar results on using deep learning to address the curse of dimensionality appear in Bauer and Kohler (2019), Schmidt-Hieber (2020), and Zhong, Mueller, and Wang (2021, 2022). However, their approach requires neural networks with sparse, bounded weights–architectures that prove difficult to construct and train, rendering them largely impractical. In contrast, we leverage recent theoretical advances in approximation error (Yarotsky 2017; Yan, Yao, and Zhou 2025) and Vapnik–Chervonenkis dimension (Bartlett et al. 2019) to adopt flexible, fully connected networks, which offer greater practical feasibility.

The rest of the article is organized as follows. In Section 2, we formulate the hypothesis testing problem and introduce the procedures for the estimation and test. In Section 3, we study the theoretical properties of the proposed estimators and tests. Section 4 presents numerical studies on simulated data and a real-data application, with the code available at *https://github.com/qxzhong/DeepSurvTest*. Concluding remarks are provided in Section 5, and two extended test statistics are included in the Appendix. Additional simulation results and mathematical proofs are provided in the supplementary material.

2. Methodology

Suppose that we record n iid copies $\mathcal{D} = \{(V_i, T_i, \Delta_i) : i = 1, \ldots, n\}$ of (V, T, Δ). Here the covariate $V = (X, Z) \in \mathbb{R}^p \times \mathbb{R}^q$ contains a p dimensional covariate X of primary interest and a q dimensional auxiliary covariate Z. The observed time T is the minimum of the event time U and the censoring time C, that is, $T = \min\{U, C\}$. The variable $\Delta = 1(U \leq C)$ is the censoring indicator, such that $\Delta = 1$ if the event time U is observed. Otherwise, $\Delta = 0$ and the censoring time C is observed.

In this article, we assume that the hazard function of the event time U for a subject with covariate $V = (X, Z)$ satisfies the proportional hazard assumption (Cox 1972, 1975):

$$\lambda(u|V) = \lambda_0(u) \exp\{g_0(V)\}, \tag{2}$$

where λ_0 is a unknown nonnegative baseline hazard function and g_0 is an unspecified function of X and Z.

The aim of this article is to test whether the primary covariate X is significant in predicting the event time U. Under the proportional hazard framework (2), this is equivalent to testing whether the function $g_0(V)$ is merely a function of the covariate Z. Thus, we consider the following test:

$$H_0 : g_0 \text{ is } X\text{-independent versus } H_1 : g_0 \text{ is } X\text{-dependent.}$$

Here g_0 is called an X-independent function if there is a function $g \in L^2(P_Z) = \{g : \mathbb{E}[g(Z)]^2 < \infty\}$ such that $g_0(V) = g(Z)$, *a.s.* P_V, where P_Z (or P_V) is the probability distributions of the random variables Z (or V). Otherwise it is called X-dependent function. Note that under the null hypothesis, we can conclude that $\mathbb{E}[1(U \leq u)|X, Z] = \mathbb{E}[1(U \leq u)|Z]$, for any $u > 0$. This is equivalent to the conditional independence of U and X given Z under the proportionality assumption in (2).

To construct the test statistic, we randomly split the data $\mathcal{D}$ into $\mathcal{K}$ disjoint subsets of equal size. The number of subsets $\mathcal{K}$ is prespecified and is typically set to 5 or 10. Without loss of generality, we assume $\tilde{n} = n/\mathcal{K}$ is an integer. We then denote the kth subset as $\mathcal{D}_k = \{(V_i^{(k)}, T_i^{(k)}, \Delta_i^{(k)}) : i = 1, \ldots, \tilde{n}\}$ and its complement as $\mathcal{D}_k^c = \{(V_i^{(-k)}, T_i^{(-k)}, \Delta_i^{(-k)}) : i = 1, \ldots, n - \tilde{n}\}$. The complement set $\mathcal{D}_k^c$ will be used to get estimators of function g_0 and $\mathcal{D}_k$ is used to construct the test based on the estimators. More specifically, denote the at risk process as $Y(t) = 1(T \geq t)$, we estimate g_0 by maximizing the logarithm of the Cox's partial likelihoods (Cox 1972),

$$\begin{aligned}\hat{g}_n^{(k)} = \mathop{\arg\max}_{g \in \mathcal{G}(q, L, \boldsymbol{p}, K)} \frac{1}{n - \tilde{n}} \sum_{i=1}^{n-\tilde{n}} \Delta_i^{(-k)} \Big[g(Z_i^{(-k)}) \\ - \log\Big(\frac{1}{n - \tilde{n}} \sum_{j=1}^{n-\tilde{n}} Y_j^{(-k)}(T_i^{(-k)}) \exp\{g(Z_j^{(-k)})\} \Big) \Big],\end{aligned} \tag{3}$$

and

$$\begin{aligned}\hat{g}_a^{(k)} = \mathop{\arg\max}_{g \in \mathcal{G}(p+q, L, \boldsymbol{p}, K)} \frac{1}{n - \tilde{n}} \sum_{i=1}^{n-\tilde{n}} \Delta_i^{(-k)} \Big[g(V_i^{(-k)}) \\ - \log\Big(\frac{1}{n - \tilde{n}} \sum_{j=1}^{n-\tilde{n}} Y_j^{(-k)}(T_i^{(-k)}) \exp\{g(V_j^{(-k)})\} \Big) \Big],\end{aligned} \tag{4}$$

under both the null and alternative hypothesis, respectively, where $\mathcal{G}(q, L, \boldsymbol{p}, K)$ and $\mathcal{G}(p + q, L, \boldsymbol{p}, K)$ are two neural network classes defined later in (5) for some integer parameters L, K, and $\boldsymbol{p}$ and $Y_j^{(-k)}(t) = 1(T_j^{(-k)} \geq t)$ is a realization of $Y(t)$. A neural network g is a sequential composite function with a linear operator followed by an elementwise nonlinear operator:

$$g(x) = \mathcal{T}_L \circ \zeta \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_1 \circ \zeta \circ \mathcal{T}_0(x),$$

where L is the number of layers of the neural network, $\mathcal{T}_l(x) = A_l x + b_l, l = 0, 1, \ldots, L$ is a linear operator with matrices $A_l \in \mathbb{R}^{p_{l+1} \times p_l}$ and vectors $b_l \in \mathbb{R}^{p_{l+1}}$, and ζ is a nonlinear operator, called the activation function, that acts on each element of a vector. There are many choices for the activation function, such as sigmoid $\zeta(x) = (1 + e^{-x})^{-1}$, hyperbolic tangent $\zeta(x) = (e^x - e^{-x})(e^x + e^{-x})^{-1}$ and rectified linear unit $\zeta(x) = \max(0, x)$. We employ the rectified linear unit (ReLU) as the activation function, which is widely used (LeCun, Bengio, and Hinton 2015; Ramachandran, Zoph, and Le 2017) due to its empirical success (Krizhevsky, Sutskever, and Hinton 2012). Let $\mathbb{Z}_+$ be the set of positive integers. For $d, L \in \mathbb{Z}_+$, $\boldsymbol{p} = (p_1, \ldots, p_L)^\top \in \mathbb{Z}_+^L$, $K > 0$, denote

$$\begin{aligned}\mathcal{G}(d, L, \boldsymbol{p}, K) = \{g : \mathbb{R}^d \to [-K, K] : g(x) \\ = \mathcal{T}_L \circ \zeta \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_1 \circ \zeta \circ \mathcal{T}_0(x) \text{ with } p_0 = d, \\ p_{L+1} = 1, \mathcal{T}_l(x) = A_l x + b_l, A_l \in \mathbb{R}^{p_{l+1} \times p_l}, \\ b_l \in \mathbb{R}^{p_{l+1}} \text{ for } l = 0, \ldots, L\}.\end{aligned} \tag{5}$$

We then construct the test statistic by

$$W_n = \frac{1}{\mathcal{K}} \sum_{k=1}^{\mathcal{K}} W_n^{(k)}, \tag{6}$$

where

$$\begin{aligned}W_n^{(k)} = \frac{1}{\tilde{n} \sigma_{\tilde{n}}^{(k)}} \sum_{i=1}^{\tilde{n}} \Delta_i^{(k)} \{ l_{\tilde{n}}(T_i^{(k)}, V_i^{(k)}, \hat{g}_a^{(k)}) \\ - l_{\tilde{n}}(T_i^{(k)}, Z_i^{(k)}, \hat{g}_n^{(k)}) \} + \rho_n e_i^{(k)},\end{aligned} \tag{7}$$

and $l_{\tilde{n}}(T_i^{(k)}, V_i^{(k)}, \hat{g}_a^{(k)}) = \hat{g}_a^{(k)}(V_i^{(k)}) - \log\big(\frac{1}{\tilde{n}} \sum_{j=1}^{\tilde{n}} Y_j^{(k)}(T_i^{(k)}) \exp\{\hat{g}_a^{(k)}(V_j^{(k)})\}\big)$, $l_{\tilde{n}}(T_i^{(k)}, Z_i^{(k)}, \hat{g}_n^{(k)}) = \hat{g}_n^{(k)}(Z_i^{(k)}) - \log\big(\frac{1}{\tilde{n}} \sum_{j=1}^{\tilde{n}} Y_j^{(k)}(T_i^{(k)}) \exp\{\hat{g}_n^{(k)}(Z_j^{(k)})\}\big)$, ρ_n is a positive number called the perturbation size, $\{e_i^{(k)} : i = 1, \ldots, \tilde{n}\}$ are iid random variables from some distribution of variance one, such as we use a standard normal distribution in this article, and

$$\sigma_{\tilde{n}}^{(k)} = \Big\{ \frac{1}{\tilde{n} - 1} \sum_{i=1}^{\tilde{n}} \Big(\Delta_i^{(k)} \{ l_{\tilde{n}}(T_i^{(k)}, V_i^{(k)}, \hat{g}_a^{(k)}) - l_{\tilde{n}}(T_i^{(k)}, Z_i^{(k)}, \hat{g}_n^{(k)}) \} - \bar{\mu}_k \Big)^2 + \rho_n^2 \Big\}^{1/2}$$

with $\bar{\mu}_k = \frac{1}{\tilde{n}}\sum_{i=1}^{\tilde{n}} \Delta_i^{(k)}\{l_{\tilde{n}}(T_i^{(k)}, V_i^{(k)}, \hat{g}_a^{(k)}) - l_{\tilde{n}}(T_i^{(k)}, Z_i^{(k)}, \hat{g}_n^{(k)})\}$. The test statistic W_n is the standardized difference of the logarithm of the Cox's partial likelihoods between the alternative and null hypothesis, plus a perturbation stochastic term to prevent the difference from asymptotically vanishing to zero under the null hypothesis. More specifically, under the null hypothesis and $\rho_n = 0$, the standard deviation $\sigma_{\tilde{n}}^{(k)}$ will converge to zero at certain rate and $\frac{1}{\tilde{n}}\sum_{i=1}^{\tilde{n}} \Delta_i^{(k)}\{l_{\tilde{n}}(T_i^{(k)}, V_i^{(k)}, \hat{g}_a^{(k)}) - l_{\tilde{n}}(T_i^{(k)}, Z_i^{(k)}, \hat{g}_n^{(k)})\}$ vanished at the same rate as $\sigma_{\tilde{n}}^{(k)}$ since the bias-variance tradeoff for training a deep learning model. This implies that $\sqrt{\tilde{n}}W_n^{(k)} \to \infty$, leading to the invalid inference and large Type I error. Here we treat ρ_n as a tuning parameter whose selection procedure is displayed in Section 4.1.

3. Theoretical Results

This section derives the asymptotic null distribution and consistency of the test statistic W_n in (6). We first introduce the rate of the neural network estimators $\hat{g}_n^k$ and $\hat{g}_a^k$, which is fundamental to testing. Without restrictions on the nonparametric function g_0, the convergence rate of its estimators would be infeasible. In this article, we assume the function g_0 belongs to some Hölder class. A Hölder class with smoothness α, domain $\mathbb{D}$ and boundedness M is defined by

$$\mathcal{H}_\alpha(\mathbb{D}, M) = \{g : \mathbb{D} \to \mathbb{R} : \sum_{\beta:|\beta|<\alpha} \|\partial^\beta g\|_\infty + \sum_{\beta:|\beta|=\lfloor\alpha\rfloor} \sup_{x,y\in\mathbb{D}, x\neq y} \frac{|\partial^\beta g(x) - \partial^\beta g(y)|}{\|x-y\|_\infty^{\alpha-\lfloor\alpha\rfloor}} \le M\},$$

where $\partial^\beta = \partial^{\beta_1}\cdots\partial^{\beta_r}$, $|\beta| = \beta_1 + \cdots + \beta_r$ with $\beta = (\beta_1, \ldots, \beta_r)^\top$, and $\lfloor\alpha\rfloor$ stands for the largest integer strictly smaller than α. This is a rich function classes that includes polynomials, exponentials and Lipschitz continuous functions.

With the notations above, we make the following assumptions.

Assumption 1. The support of the covariate $V = (X, Z)$ is on a compact set $\mathcal{V}$ of $\mathbb{R}^{p+q}$ with fixed p and q, and without loss of generality, we take $\mathcal{V} = [0,1]^{p+q}$, and assume that the joint probability density function of X and Z is bounded away from zero.

Assumption 2. The function g_0 is an element of a Hölder class $\{g \in \mathcal{H}_\alpha([0,1]^{p+q}, M) : \mathbb{E}\{g_0(V)\} = 0\}$ for some $\alpha > 0, M > 0$.

Assumption 3. Conditional on the covariate V, the event time U and censoring time C are independent.

Assumption 4. There exists a constant $c > 0$, such that $P(\Delta = 1|V) > c$ and $P(T \ge \tau|V) \ge c$ are valid almost surely with respect to the probability of V, where τ denotes the study's end time.

Assumption 5. The neural network structure used for estimation in (3) and (4) satisfies $K > 2M$ and $\tilde{p}L = O(n^{q/(4\alpha+2q)}(\log n)^2)$ under the null hypothesis H_0, or $\tilde{p}L = O(n^{(p+q)/(4\alpha+2p+2q)}(\log n)^2)$ under the alternative hypothesis H_1, where $\tilde{p} := \max\{p_1, \ldots, p_L\}$.

Assumption 6. There exists a constant $\rho > 0$, such that $\rho_n \to \rho$ as $n \to \infty$.

Assumptions 1 and 2 are standard in nonparametric estimation (van der Vaart and Wellner 1996; van der Vaart 2000). The zero expectation of g_0 in Assumption 2 is for identifiability, and the intercept term is absorbed into the baseline hazard λ_0. Assumptions 3 and 4 are commonly assumed in survival analysis (Andersen and Gill 1982; Murphy, Rossini, and van der Vaart 1997), whose $P(T \ge \tau|V) \ge c$ along with the boundedness of g_0 prevent the baseline cumulative hazard $\Lambda_0(t) = \int_0^t \lambda_0(s)ds$ to be infinite at time τ. Assumption 5 make constraints on the configuration of the neural network in order to establish the convergence rate of its estimators, while Assumption 6 on perturbation size ρ_n ensures the asymptotic normality and consistency of the test statistic.

Note that if $\hat{g}_n^{(k)}(\cdot)$ (or $\hat{g}_a^{(k)}(\cdot)$) is a maximum partial likelihood estimator in (3) (or (4)), any constant shift, that is, $\hat{g}_n^{(k)}(\cdot) + c$ (or $\hat{g}_a^{(k)}(\cdot) + c$) is also a maximum partial likelihood estimator. Thus, we assume that $1/\tilde{n}\sum_{i=1}^{\tilde{n}} \hat{g}_n^{(k)}(Z_i^{(k)}) = 0$ and $1/\tilde{n}\sum_{i=1}^{\tilde{n}} \hat{g}_a^{(k)}(V_i^{(k)}) = 0$. These can be realized by using $\hat{g}_n^{(k)}(z) - 1/\tilde{n}\sum_{i=1}^{\tilde{n}} \hat{g}_n^{(k)}(Z_i^{(k)})$ and $\hat{g}_a^{(k)}(z) - 1/\tilde{n}\sum_{i=1}^{\tilde{n}} \hat{g}_a^{(k)}(Z_i^{(k)})$ as estimators when $\hat{g}_n^{(k)}(\cdot)$ and $\hat{g}_a^{(k)}(\cdot)$ are derived from (3) and (4), respectively.

Theorem 3.1. Suppose that Assumptions 1–5 hold. Under the null hypothesis H_0, for $k = 1, \ldots, \mathcal{K}$,

$$\|\hat{g}_n^{(k)} - g_0\|_{L^2([0,1]^q)} \asymp \|\hat{g}_a^{(k)} - g_0\|_{L^2([0,1]^{p+q})} = O_p\Big(\frac{\log^2 n}{n^{\alpha/(2\alpha+q)}}\Big), \tag{8}$$

where $\|\hat{g}_a^{(k)} - g_0\|^2_{L^2([0,1]^{p+q})} = \int_{[0,1]^p}\int_{[0,1]^q}(\hat{g}_a^{(k)}(x,z) - g_0(z)1(x \in [0,1]^p))^2 dzdx$. Under the alternative hypothesis H_1,

$$\|\hat{g}_a^{(k)} - g_0\|_{L^2([0,1]^{p+q})} = O_p\Big(\frac{\log^2 n}{n^{\alpha/(2\alpha+p+q)}}\Big). \tag{9}$$

The convergence rates in Theorem 3.1 involves the errors of statistical estimation and neural network approximation, which is the distance between the neural network space $\mathcal{G}(q, L, \boldsymbol{p}, K)$ (or $\mathcal{G}(p+q, L, \boldsymbol{p}, K)$) and the function g_0. Intuitively, a more complex neural network class is able to approximate a function more accurately (Yarotsky 2017; Gühring, Kutyniok, and Petersen 2020), but this is accomplished at a cost to increase the estimation variance (Anthony and Bartlett 1999). This is the usual tradeoff between estimation errors and approximation errors, as reflected in the proof of Theorem 3.1.

It should be noted that the estimators $\hat{g}_n^{(k)}$ and $\hat{g}_a^{(k)}$ share the same convergence rate in (8) under the null hypothesis H_0, which may seem counterintuitive given that the input dimension of $\hat{g}_a^{(k)}$ (i.e., $p+q$) is larger than that of $\hat{g}_n^{(k)}$ (i.e., q), the dimension of the true function g_0. However, it is well known that deep learning can effectively learn the low-dimensional structure of a function embedded in a high-dimensional space. For example, the true function g_0 under the null hypothesis is a q-dimension function within a $(p+q)$-dimensional space. Such adaptability allows neural networks to avoid the curse of dimensionality (Bauer and Kohler 2019; Schmidt-Hieber 2020). We formalize this result within the survival analysis framework. See more general cases below.

Next, we further demonstrate that neural networks can learn the low-dimensional structure of data embedded in a high-dimensional space. We assumes that g_0 is a member of some composite functional class:

$$\begin{aligned}\mathcal{H}(\bar{K}, \boldsymbol{\alpha}, \boldsymbol{d}, \tilde{\boldsymbol{d}}, M) = \{g = \boldsymbol{g}_{\bar{K}} \circ \cdots \circ \boldsymbol{g}_1 : \boldsymbol{g}_k = (g_{k1}, \ldots, g_{kd_{k+1}})^\top, \\ g_{kl} \in \mathcal{H}_{\alpha_i}([a_k, b_k]^{\tilde{d}_k}, M), \text{for some } |a_k|, |b_k| \le M \\ \textit{and } d_1 = p + q, d_{\bar{K}+1} = 1, \mathbb{E}\{g_0(V)\} = 0\},\end{aligned}$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{\bar{K}})^\top \in \mathbb{R}_+^{\bar{K}}, \boldsymbol{d} = (d_1, \ldots, d_{\bar{K}+1})^\top \in \mathbb{N}^{\bar{K}+1}$ and $\tilde{\boldsymbol{d}} = (\tilde{d}_1, \ldots, \tilde{d}_{\bar{K}})^\top \in \mathbb{N}^{\bar{K}}$. Usually the *intrinsic dimension* $\tilde{\boldsymbol{d}}$ (Schmidt-Hieber 2020; Zhong, Mueller, and Wang 2022) is much smaller than the input covariate dimension $\boldsymbol{d}$, that is, $\tilde{d}_k \le d_k$ for $k = 1, \ldots, \bar{K}$. Denote $\tilde{\alpha}_k = \alpha_k \prod_{j=k+1}^{\bar{K}} (\alpha_j \wedge 1)$ and $r_n = \max_{k=1,\ldots,\bar{K}} n^{-\tilde{\alpha}_k/(2\tilde{\alpha}_k + \tilde{d}_k)}$ with notation $a \wedge b := \min\{a, b\}$, we have the following corollary.

Corollary 3.1. Suppose that Assumptions 1, 3, and 4 hold. If $g_0 \in \mathcal{H}(\bar{K}, \boldsymbol{\alpha}, \boldsymbol{d}, \tilde{\boldsymbol{d}}, M)$ and neural network structure in (5) satisfies $K > 2M, \tilde{p}L = O(n^{1/2} r_n (\log n)^2)$. Then under the null hypothesis H_0, for each $k = 1, \ldots, \mathcal{K}$,

$$\|\hat{g}_n^{(k)} - g_0\|_{L^2([0,1]^q)} \asymp \|\hat{g}_a^{(k)} - g_0\|_{L^2([0,1]^{p+q})} = O_p(r_n \log^2 n). \tag{10}$$

Under the alternative hypothesis H_1,

$$\|\hat{g}_a^{(k)} - g_0\|_{L^2([0,1]^{p+q})} = O_p(r_n \log^2 n). \tag{11}$$

Corollary 3.1 shows that the convergence rates are adaptive to the intrinsic dimension of g_0 when it has a low dimensional structure embedded in the high dimension space. That is, the proposed deep learning approach is able to detect the unknown low dimensional structure of data, and thus circumvents the curse of dimensionality that traditional nonparametric smoothing methods suffer. For example, when

$$\begin{aligned} g_0(v) = g_{31}(g_{21}(g_{11}(z_1, z_2), g_{12}(z_3, z_4), g_{13}(z_5, z_6)), \\ g_{22}(g_{13}(z_7, z_8), g_{14}(z_9, z_{10}), g_{15}(z_{11}, z_{12}))) \end{aligned}$$

and g_{ij} are three times continuously differentiable, then the resulting rates in (10) are of the order $n^{-1/3} \log^2 n$ rather than the slower convergence rate of order $n^{-1/6}$ by traditional smoothing methods, such as B-spline regression and kernel smoothing.

Theorem 3.2. Suppose that Assumptions 1–6 hold. If $q - 2\alpha < 0$, then under the null hypothesis H_0,

$$\sqrt{n} W_n \xrightarrow{d} N(0, 1). \tag{12}$$

Note that conditional on $\mathcal{D}_k^c$ (or $\hat{g}_n^{(k)}$ and $\hat{g}_a^{(k)}$), the kth term $W_n^{(k)}$ of the test W_n defined in (6) is not the sum of independent terms due to the presence of the at risk process, nor is W_n. For the proof of this key theorem, we decompose $\sqrt{n} W_n$ into two terms, where we show that, one term is a sum of iid variables and thus converges in distribution to a standard normal distribution, while the other term converges weakly to zero. The regularity condition $q - 2\alpha < 0$ in this theorem is standard in semi-parametric estimation (Bickel et al. 1993; van der Vaart 2000; Kosorok 2008) and the cross-fitting literature (Chernozhukov et al. 2018). It requires that the nuisance function estimator converges at a rate slightly faster than order $n^{-1/4}$ to ensure root-n consistency of the target estimator. This condition is satisfied when the nuisance function is sufficiently smooth

Once the weak convergence of $\sqrt{n} W_n$ is established, the asymptotic normality allows us to test the null hypothesis with some ρ_n. We treat ρ_n in this article as a tuning parameter, whose choice is discussed in Section 4.1.

Next we demonstrate the consistency of the test under the alternative hypothesis H_1.

Theorem 3.3. Suppose that Assumptions 1–6 and $p + q - 2\alpha < 0$ hold. Under the alternative hypothesis H_1, if there exist $\epsilon > 0$, such that $\inf_{g \in L^2(P_Z)} \|g - g_0\|_{L^2([0,1]^{p+q})} \ge \epsilon$, then for every $c > 0$,

$$\lim_{n \to \infty} P(\sqrt{n} |W_n| \ge c) = 1. \tag{13}$$

In Theorem 3.2, we assume that the dimension and smoothness of g_0 satisfy $q - 2\alpha < 0$, or in Theorem 3.3, $p + q - 2\alpha < 0$, to ensure the neural network estimators converge fast enough. This convergence leads to certain terms in the proofs becoming asymptotically negligible, resulting in the test statistic being asymptotically normal or consistent. Although these assumptions may fail if the structure of the function class in which g_0 resides is quite complex, this challenge can be overcome if g_0 has a low-dimensional structure. Below are the asymptotic normality and consistency results of the test corresponding to Corollary 3.1, where the function g_0 has a low-dimensional structure.

Corollary 3.2. Suppose that Assumptions 1, 3, 4 and 6 hold. If $g_0 \in \mathcal{H}(K, \boldsymbol{\alpha}, \boldsymbol{d}, \tilde{\boldsymbol{d}}, M)$, $\tilde{d}_k - 2\tilde{\alpha}_k < 0$ for $k = 1, \ldots, \bar{K}$ and the neural network structure satisfies $K > 2M$ and $\tilde{p}L = O(n^{1/2} r_n (\log n)^2)$. Then under the null hypothesis H_0,

$$\sqrt{n} W_n \xrightarrow{d} N(0, 1).$$

Under the alternative hypothesis H_1, if there exist $\epsilon > 0$, such that $\inf_{g \in L^2(P_Z)} \|g - g_0\|_{L^2([0,1]^{p+q})} \ge \epsilon$, then for every $c > 0$,

$$\lim_{n \to \infty} P(\sqrt{n} |W_n| \ge c) = 1.$$

4. Numerical Studies

In this section, we examine the finite sample performance of the proposed method. All models to obtain the partial likelihood estimates in our simulations and data analyses were implemented using the PyTorch library (Paszke et al. 2019).

4.1. Experimental Setup

Stochastic gradient descent optimization is highly scalable and effective for function estimation via neural networks in practice, despite early concerns about nonconvexity (Robbins and Monro 1951). In this article, we use the *Adam* optimizer (Kingma and

Ba 2014), an adaptive stochastic gradient method, to update the weights, A_l and b_l, of the neural networks in (5). These weights are randomly initialized in the traditional manner. Additional tuning parameters, such as the number of layers (L) and the number of neurons (p_l), were optimized via grid search: we selected values that maximize the partial likelihood on a 20% holdout validation set from the estimation datasets. The learning rate (Goodfellow, Bengio, and Courville 2016) was set automatically using *pycox* (Kvamme, Borgan, and Scheel 2019).

To perform the inference, we need to specify the perturbation size ρ_n. As suggested by Dai, Shen, and Pan (2022), we first obtain a permuted sample $\tilde{X}_1, \dots, \tilde{X}_n$ from $X_1, \dots, X_n$. Then we get the estimates of g_0 under both the null and alternative hypotheses with respect to the "new" sample $\{(\tilde{X}_i, Z_i, T_i, \Delta_i) : i = 1, \dots, n\}$ and calculate the corresponding p-value with the given $\tilde{\rho}$. This procedure is repeated $\mathcal{N}$ times to get $\mathcal{N}$ p-values, denoted as $p_1(\tilde{\rho}), \dots, p_{\mathcal{N}}(\tilde{\rho})$. With a nominal level κ, the perturbation size ρ_n are obtained as

$$\rho_n = \min\{\tilde{\rho} \in \mathcal{A} : \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} 1(p_i(\tilde{\rho}) \le \kappa) \le \kappa\}, \tag{14}$$

where $\mathcal{A}$ are the sets of grid points for $\tilde{\rho}$.

4.2. Simulations

We first generate $\tilde{Z} = (\tilde{Z}_1, \dots, \tilde{Z}_{10})$ from a Gaussian copula on $[0, 2]$ with correlation parameter 0.5. Marginally, each coordinate $\tilde{Z}_i, i = 1, \dots, 10$ is uniformly distributed on $[0, 2]$ and correlation coefficient between two coordinates is 0.5. we set the covariate $Z = (\tilde{Z}_1, \dots, \tilde{Z}_9)$ and the covariate X to be $X = \tilde{Z}_{10}$.

Conditional on the covariate (X, Z), the event time U is generated from the Weibull distribution

$$S(u|X, Z) = \exp\left[-0.2u \exp\{g_0(X, Z; \gamma)\}\right], \tag{15}$$

where $g_0(x, z; \gamma) = g_1(z) + \gamma g_2(x, z)$, for which

$$\begin{aligned} g_1(z) = {}& z_1^3 + \log(z_2 + 1) z_5 z_6 + \exp(z_4/4)(z_3 z_4 + 1)^{1/2} \\ & + \frac{z_7\{\exp(z_8 - 1) - \exp(1 - z_8)\}}{\exp(z_8 - 1) + \exp(1 - z_8)} + z_9^2 - c_1, \end{aligned} \tag{16}$$

and $g_2(x, z) = 0.2 z_1 (x - 1)^2 + 3(\sin(z_2 - 1) + z_3^2)(x - 1) + \log(z_4 \cos(\pi z_5) + 3) + z_6 z_7^3/3 + (z_8 - 1)^2 + \exp(z_8 z_9/2) - c_2$. Here c_1, c_2 satisfying $\mathbb{E} g_1(Z) = \mathbb{E} g_2(X, Z) = 0$. The factor γ is set to be $0.0, 0.2, 0.4, 0.6, 0.8, 1.0$, respectively. Here the factor γ controls the distance between the function spaces of the null and alternative hypotheses. When $\gamma = 0$, the true function $g_0(X, Z; \gamma)$ falls in the space of the null hypothesis, otherwise $g_0(X, Z; \gamma)$ is in the space of the alternative hypothesis. The distance between $g_0(X, Z; \gamma)$ and the null space becomes larger as γ increases. For the censoring time C, we generate it from an exponential distribution with mean $\mu = 10.0$. The final observations are $(V_i, T_i, \Delta_i), i = 1, \dots, n$, where $V_i = (X_i, Z_i)$, $T_i = \min\{U_i, C_i\}$, $\Delta_i = 1(U_i \le T_i)$ and $n = 400, 600, 800$.

We follow the implementation in Section 4.1 to select the tuning parameters of neural network based on several trial runs. The resulting values are $L = 1, p_l = 32$. The procedure for selecting perturbation size ρ_n in (14) is based on $\mathcal{N} = 200$ times of permutation among candidate $\mathcal{A} = \{10^{-3+i/50} : i = 0, 1, \dots, 200\}$.

For each run, we random split the simulated data into $\mathcal{K} = 10$ parts of equal size. With $\hat{g}^{(k)}, k = 1, \dots, \mathcal{K}$ being the estimates of the function g_0, we define the relative mean squared error (RMSE) as

$$\text{RMSE} = \frac{\frac{1}{\mathcal{K}} \sum_{k=1}^{\mathcal{K}} \frac{1}{\tilde{n}} \sum_{i=1}^{\tilde{n}} \{\hat{g}^{(k)}(V_i^{(k)}) - g_0(V_i^{(k)}) - (\bar{\hat{g}}^{(k)} - \bar{g}_0^{(k)})\}^2}{\frac{1}{\mathcal{K}} \sum_{k=1}^{\mathcal{K}} \frac{1}{\tilde{n}} \sum_{i=1}^{\tilde{n}} \{g_0(V_i^{(k)}) - \bar{g}_0^{(k)}\}^2}, \tag{17}$$

where $\{V_i^{(k)} : i = 1, \dots, \tilde{n}\}$ are the covariates in the kth subset and $\bar{\hat{g}}^{(k)} = 1/\tilde{n} \sum_{i=1}^{\tilde{n}} \hat{g}^{(k)}(V_i^{(k)})$, $\bar{g}_0^{(k)} = 1/\tilde{n} \sum_{i=1}^{\tilde{n}} g_0(V_i^{(k)})$.

Based on 200 simulations, Table 1 shows the average of the RMSEs for the estimates $\hat{g}_n^{(k)}$ and $\hat{g}_a^{(k)}$ of g_0. In general, the average RMSEs decrease steadily as the sample size n increases from 400 to 800 when γ is fixed. The performance deteriorates as γ gets larger when sample size n is fixed. This is because the variance gets larger as γ gets larger. Overall, the simulation results are in agreement with Theorem 3.1. That is, the RMSEs results of $\hat{g}_n^{(k)}$ and $\hat{g}_a^{(k)}$ are similar under the null hypotheses (i.e., $\gamma = 0$), while $\hat{g}_n^{(k)}$ performs clearly worse than $\hat{g}_a^{(k)}$ under the alternative hypotheses, especially when the distance between the function $g_0(X, Z; \gamma)$ and the null space is large (i.e., γ is large). Moreover, we conducted experiments to demonstrate that neural networks are capable of detecting the intrinsic dimension of the function g_0. With the true function $g_0 = g_1$ (i.e., $\gamma = 0$), whose intrinsic dimension is 5, we set the dimension of input covariates to be 10, 20, and 30, respectively, and obtained their neural network estimates with sample sizes of $n = 400, 600, 800, 1000, 1200, 1400$. Here we generate all inputs from a Gaussian copula on $[0, 2]$ with a correlation parameter of 0.5. We record the RMSE of the estimates and plotted the logarithm of the sample size n against the logarithm of the RMSE in Figure 1. It is worth noting that Corollary 3.1 suggests that the convergence rates of neural network estimates are proportional to $n^{-\alpha}$ for some constant α. The slope of the straight line in the log-log plot provides an estimate of α. We observed that the three lines in Figure 1 are nearly parallel, indicating that neural network estimates with different input dimensions exhibit a similar convergence rate. This further implies that neural networks are able to adapt the low-dimensional structure of a function embedded in a high-dimensional space, thus, overcoming the curse of the dimensionality.

Furthermore, we record the rejection probability among the 200 simulations at nominal significant level 0.05 in Table 2. It shows that the Type I error probabilities (under $\gamma = 0.0$) are approximately equal to the nominal significant level 0.05 and

Table 1. Relative mean squared error for the estimates of g_0 under null and alternative hypotheses.

n	RMSE	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
400	RMSE_a	0.088	0.115	0.160	0.198	0.235	0.268
	RMSE_n	0.085	0.132	0.226	0.328	0.398	0.462
600	RMSE_a	0.068	0.096	0.142	0.177	0.218	0.243
	RMSE_n	0.063	0.112	0.215	0.328	0.404	0.470
800	RMSE_a	0.059	0.083	0.125	0.166	0.199	0.227
	RMSE_n	0.058	0.104	0.205	0.303	0.389	0.454

NOTE: RMSE_a: RMSE for $\hat{g}_a^{(k)}$; RMSE_n: RMSE for $\hat{g}_n^{(k)}$.

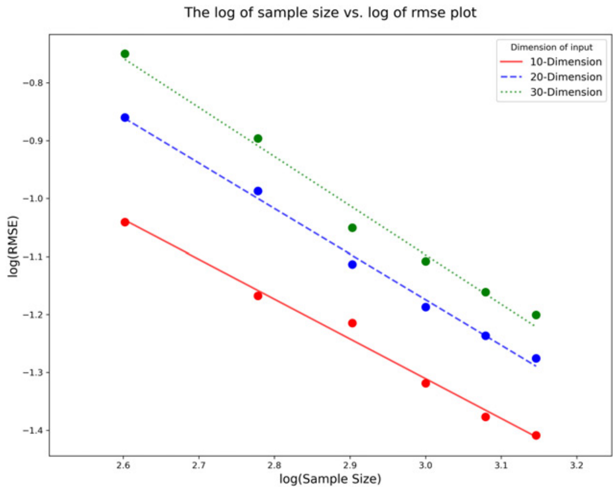

Figure 1. The log of sample size versus log of rmse plot.

Table 2. Rejection probability at the nominal significant level 0.05.

n	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
400	0.060	0.345	0.865	0.990	1.000	1.000
600	0.055	0.595	0.980	0.995	1.000	1.000
800	0.045	0.835	1.000	1.000	1.000	1.000

the power increases as the sample size or γ increasing. Table 7 in the supplementary material displays rejection probabilities for input covariates with dimensions of 20 and 30, showing results consistent with those for 10 dimensions. We also conduct an experiment to analyze the sensitivity of varying perturbation sizes ρ_n to the rejection probability. Table 3 displays the rejection probabilities at nominal significant level 0.05 with $\rho_n \in \{10^{-3}, 10^{-2.5}, 10^{-2.0}, 10^{-1.5}, 10^{-1.0}, 10^{-0.5}\}$. It shows that small perturbation sizes tend to result in larger Type I errors and smaller Type II errors, while the opposite is observed for large perturbation sizes. The results remain relatively stable with smaller perturbation sizes. It should be noted that the Type I error under the null hypothesis should be asymptotically independent of the perturbation size ρ_n. However, the Type I errors display different with the varying perturbation sizes. We believe that this is because the sample sizes are not large enough relative to the likelihood difference in the test statistic. Additionally, with the number of nodes in each layer set to $p_l = 32$, we record the rejection probabilities for neural networks with 1–3 layers in Table 4. The results show that all configurations control the Type I error while a larger number of layers (2 or 3) yields worse statistical power compared to the optimal single-layer network.

Next, we compare the proposed method with three alternatives: (a) the conventional Cox model (Cox 1972), (b) the single-splitting approach, and (c) the multiple-splitting approach. The introduction of the latter two methods with their theoretical justification are provided in Appendix. Similarly, we first generate $\tilde{Z} = (\tilde{Z}_1, \ldots, \tilde{Z}_{10})$ from a Gaussian copula on $[-1, 1]$ with correlation parameter 0.1 and set the covariate $Z = (\tilde{Z}_1, \ldots, \tilde{Z}_9)$ and the covariate X to be $X = \tilde{Z}_{10}$. We generate the event time U from a survival function defined as $S(u|X, Z) = \exp[-0.2u \exp\{g_0(X, Z; \gamma)\}]$. Here $g_0(x, z; \gamma) = g_1(z) + \gamma g_3(x, z)$, where g_1 is defined in (16) and $g_3(x, z) = 2x^2 - c_3$, with the constant c_3 such that $\mathbb{E}g_3(X, Z) = 0$. The censoring time C follows an exponential distribution with mean 10.0.

We first compare the RMSE results with those for the estimates of g_0 from the conventional Cox model (Cox 1972) in Table 5. The proposed CF method consistently demonstrates superior performance with lower MSE compared to the conventional Cox model. Notably, the RMSE from CF decreases as the sample size increases, whereas the Cox model fails to show comparable improvement with increasing sample sizes. Table 6 compares the rejection probabilities at the 0.05 significance level for four methods: the conventional Cox model (Cox), the proposed cross-fitting method (CF), and the single- and multiple-splitting approaches. Under the null hypothesis ($\gamma = 0$), CF, Cox and single-splitting maintain the nominal levels across different sample sizes, while the multiple-splitting approach are consistently conservative (type one errors are 0.000). Under the alternative hypothesis ($\gamma > 0$), CF demonstrates superior performance, with its power increasing with both γ and n. In contrast, Cox and single-splitting show only modest power gains, while multiple-splitting exhibits zero power across most scenarios.

In summary, these simulation results highlight the favorable performance of the proposed CF method. Compared to alternative approaches, CF maintains proper Type I error control while achieving substantially higher power, particularly at larger effect sizes γ and sample sizes n.

4.3. *Application to FLCHAIN Survival Data*

In this section, we apply the proposed test to the data of Assay Of Serum Free Light Chain (FLCHAIN) (Kyle et al. 2006), which is available at the R package *survival* (*https://cran.r-project.org/web/packages/survival/*). Along with the event time, defined as days from enrollment until death, and censoring indicator, six covariates (i.e., age, sex, κ serum free light chain (FLC), λ FLC, serum creatinine and indicator for diagnosis with monoclonal gammopathy) of the data are included in our analysis. After removing subjects with missing values for each feature, there are 6524 observations remaining, of which 30.1% experienced death during the study.

In this study, our interest lies in determining whether nonclonal elevation of serum FLCs, measured as the sum of κ and λ FLC (Σ FLC) (Dispenzieri et al. 2012), is associated with overall survival in the general population. Thus, we encode Σ FLC as the covariate X, and the remaining covariates as a vector variable Z. We preprocessed the data by standardizing each covariate to mean 0 and standard deviation 1. Using $\mathcal{K} = 10$ folds, we followed the method outlined in Section 4.1 to select the perturbation size ρ_n from a candidate set $\{10^{-3+i/50} : i = 0, 1, \ldots, 200\}$ based on $\mathcal{N} = 200$ permutations. To choose the tuning parameters for neural networks, we randomly split the estimation set into training and validation sets in a ratio of 80:20, where the optimal tuning parameters maximize the partial likelihood on the validation set. We also use *Adam* algorithm (Kingma and Ba 2014) to optimize the objective function with a *PyTorch's* default random initialization (Paszke et al. 2019). The resulting neural network structure is $L = 2$ and $p_l = 128$, and the perturbation size ρ_n equal to $10^{-2.96}$. With these tuning

Table 3. Rejection probability at the nominal significant level 0.05 with different perturbation size.

	$n = 400$						$n = 600$					
ρ_n	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
10^{-3}	0.060	0.345	0.865	0.990	1.000	1.000	0.055	0.595	0.980	0.995	1.000	1.000
$10^{-2.5}$	0.060	0.340	0.865	0.990	1.000	1.000	0.055	0.595	0.985	0.995	1.000	1.000
10^{-2}	0.060	0.320	0.860	0.985	0.985	1.000	0.050	0.470	0.975	0.995	1.000	1.000
$10^{-1.5}$	0.030	0.210	0.760	0.950	0.985	1.000	0.035	0.205	0.865	0.980	0.995	1.000
10^{-1}	0.010	0.105	0.340	0.520	0.720	0.825	0.040	0.095	0.270	0.525	0.735	0.820
$10^{-0.5}$	0.015	0.060	0.135	0.155	0.200	0.235	0.035	0.070	0.090	0.175	0.210	0.230

Table 4. Rejection probability at the nominal significant level 0.05 with different number of neural network layers.

	$n = 400$						$n = 600$					
# of layers	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
1	0.035	0.340	0.825	0.980	0.995	1.000	0.065	0.415	0.970	1.000	1.000	1.000
2	0.050	0.080	0.195	0.380	0.500	0.565	0.055	0.110	0.160	0.245	0.385	0.605
3	0.055	0.055	0.155	0.215	0.180	0.220	0.060	0.090	0.115	0.170	0.255	0.295

Table 5. Relative mean squared error for the estimates of g_0.

n	Method	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
400	CF	0.131	0.134	0.127	0.119	0.123	0.117
	Cox	0.142	0.138	0.132	0.130	0.127	0.126
600	CF	0.095	0.098	0.091	0.091	0.091	0.088
	Cox	0.140	0.137	0.132	0.131	0.126	0.125

NOTE: CF: estimate $\hat{g}_a^{(k)}$ from the proposed method; Cox: conventional Cox model.

Table 6. Rejection probability at the nominal significant level 0.05 for four methods.

n	Method	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
400	CF	0.055	0.045	0.075	0.090	0.105	0.175
	Cox	0.055	0.050	0.040	0.065	0.050	0.065
	Single	0.045	0.060	0.060	0.085	0.095	0.115
	Multi	0.000	0.000	0.000	0.000	0.000	0.000
600	CF	0.055	0.065	0.080	0.135	0.180	0.460
	Cox	0.070	0.045	0.040	0.050	0.035	0.055
	Single	0.060	0.060	0.070	0.100	0.130	0.170
	Multi	0.000	0.000	0.000	0.000	0.000	0.000

NOTE: CF: the proposed cross-fitting method; Cox: conventional Cox model; Single: single-splitting method; Multi: multiple-splitting method.

parameters, we implement the hypothesis testing and get a p-value of 0.00341. Therefore, we reject the null hypothesis that nonclonal elevation of serum FLCs is not associated with overall survival. Our results consistent with Gudowska-Sawczuk and Mroczko (2023) that FLC is a biomarker for some diseases.

5. Concluding Remarks

While deep learning has been highly successful in learning survival data, existing approaches have mainly focused on prediction and are not well-equipped to provide statistical inference results such as hypothesis testing. To address the challenge, This article proposes a new test based on deep learning for right censored survival data under a fully nonparametric Cox model. The proposed method employs neural networks to estimate the multivariate function g_0 and deploy data splitting with cross-fitting strategy to develop the test statistic. We provide the convergence rates of the neural network estimates, which shows that the proposed deep learning approach is able to mitigate the curse of dimensionality for large number of covariates. The consistency and asymptotic normality of the test statistic is further established to support the significance test of the covariates of interest. To the best of our knowledge, this is the first test of significance for censored survival data that employs a deep learning approach. In this implementation, the use of cross-fitting is time-consuming. One can accelerate it by parallelizing the estimation across different folds or subsets of the data. Modern computational frameworks (e.g., Spark, Dask, or GPU-accelerated libraries) can significantly reduce runing time.

Although we have only presented theoretical results for the continuous covariate V in Assumption 1, the proposed approach can be readily applied to cases where V contains discrete variables. For example, if $v = (v_1, v_2^\top)^\top$ with $v_1 \in \{0, 1\}$ being discrete and $v_2 \in [0, 1]^{p+q-1}$ being continuous, the true function $g_0(v)$ can be represented as $g_0(v) = 1(v_1 = 0)g_1(v_2) + 1(v_1 = 1)g_2(v_2)$. The neural network estimate $\hat{g}(v) = 1(v_1 = 0)\hat{g}_1(v_2) + 1(v_1 = 1)\hat{g}_2(v_2)$, where $\hat{g}$ is a neural network with $\hat{g}_1$ and $\hat{g}_2$ being two sub-neural networks with the same input v_2 without sharing weights.

Several extensions are worth pursuing in the future. First, we merely investigate a significance test on survival data under the framework of Cox model (Cox 1972). It will be of interest to further consider more general models, such as the extended hazard model (Ciampi and Etezadi-Amoli 1985; Zhong, Mueller, and Wang 2021) or nonparametric survival models (Beran 1981) to avoid model misspecification. Second, while the current work considers only time-invariant covariates, the proposed approach could be extended to survival data with time-varying or functional covariates through the joint modelling approach (Tsiatis, Degruttola, and Wulfsohn 1995; Tseng, Hsieh, and Wang 2005; Tseng et al. 2015) and the functional Cox model (Qu, Wang, and Wang 2016; Hao et al. 2021). Third, in principle the proposed methodology can be extended to assess goodness-of-fit for specific models, such as the proportional hazards and accelerated failure time models. This extension involves estimating the corresponding functions for both a small model under the null hypothesis and the full model, which is much larger. Subsequently, a similar test statistic can be constructed using (6) to conduct a goodness-of-fit test. However, the selection of the perturbation size ρ_n is tricky, as the permuted sample approach which we employed may not be suitable for this goodness-of-fit test. Finally, while this article focuses on variable significance testing, the underlying hypothesis is equivalent to testing the conditional independence null $U \perp X \mid Z$. Conditional independence testing for fully observed outcomes has attracted much attention in recent years (Fan, Feng, and Xia 2020; Shah and Peters 2020; Dai, Shen, and Pan 2022; Liu et al. 2022; Williamson et al. 2023; Lundborg et al. 2024), yet remains understudied for censored survival data. Our work fills this gap. Future research could extend the conditional independence testing for survival data to a fully nonparametric survival model, relaxing the proportionality assumption in (2).

Appendix: Single- and Multiple-Splitting Test Statistics

We construct the test statistic W_n in (6) of the article using the cross-fitting methodby averaging the $\mathcal{K}$ test statistics:

$$W_n = \frac{1}{\mathcal{K}} \sum_{k=1}^{\mathcal{K}} W_n^{(k)}.$$

We refer to each $W_n^{(k)}$ as a single-splitting test statistic and establish the following theoretical results.

Theorem A.1. Suppose that Assumptions 1, 3, 4, and 6 hold. We assume that $g_0 \in \mathcal{H}(K, \boldsymbol{\alpha}, \boldsymbol{d}, \tilde{\boldsymbol{d}}, M)$, $r_n n^{1/4} \log^2 n \to 0$ and the neural network structure satisfies $K > 2M$ and $\tilde{p}L = O(n^{1/2} r_n (\log n)^2)$. Then under the null hypothesis H_0,

$$\sqrt{\tilde{n}} W_n^{(k)} \xrightarrow{d} N(0, 1),$$

where $\tilde{n} = [n/\mathcal{K}]$. Under the alternative hypothesis H_1, if there exist $\epsilon > 0$, such that $\inf_{g \in L^2(P_Z)} \|g - g_0\|_{L^2([0,1]^{p+q})} \geq \epsilon$, then for every $c > 0$,

$$\lim_{n\to\infty} P(\sqrt{\tilde{n}} |W_n^{(k)}| \geq c) = 1.$$

Theorem A.1 establishes that the single-split test statistic $W_n^{(k)}$ achieves proper Type I error control under the null hypothesis while maintaining consistency under alternatives. However, this approach suffers from reduced statistical power compared to the proposed cross-fitting procedure, as it uses only a subset of available data for inference.

Next, we introduce the multiple-splitting test statistic. First, we randomly split the sample into two parts with ratio ξ:1. Similar to the construction of $W_n^{(1)}$, we use this split to compute a test statistic $\tilde{W}_n^{(1)}$. Repeating this process B times yields B independent test statistics $\tilde{W}_n^{(b)}, b = 1, \ldots, B$. The multiple-splitting test statistic is then defined as

$$\tilde{W}_n = \frac{1}{B} \sum_{b=1}^{B} \tilde{W}_n^{(b)}.$$

We then establish a property on $\tilde{W}_n$ based on Lundborg et al. (2024) and Theorem A.1.

Theorem A.2. Suppose that Assumptions 1, 3, 4, and 6 hold. We assume that $g_0 \in \mathcal{H}(K, \boldsymbol{\alpha}, \boldsymbol{d}, \tilde{\boldsymbol{d}}, M)$, $r_n n^{1/4} \log^2 n \to 0$ and the neural network structure satisfies $K > 2M$ and $\tilde{p}L = O(n^{1/2} r_n (\log n)^2)$. Then under the null hypothesis H_0, for a fixed ξ and any given $\alpha \in (0, 1)$, we have

$$\lim_{n\to\infty} P\Big(\sqrt{n}\tilde{W}_n \geq \frac{\sqrt{\xi+1}\varphi(z_{1-\alpha})}{\alpha}\Big) \leq \alpha,$$

where φ is the density function of $N(0, 1)$ and $z_{1-\alpha}$ the critical value from $N(0, 1)$ corresponding to a significance level of α.

From Theorem A.2, we can choose $\sqrt{\xi+1}\varphi(z_{1-\alpha})/\alpha$ as a threshold for the multiple-splitting test statistic $\tilde{W}_n$ to control Type I error of at most α. For example, it becomes $2.06\sqrt{\xi+1}$ when $\alpha = 0.05$.

Supplementary Materials

The supplementary material provides all technical proofs and additional simulation results.

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